PHYS 4343 FS26: Atomic Physics

Exam 2

Name: _____

Instructions

- **Time:** 60 minutes.
- This is a **closed-book and closed-notes** exam.
- The exam contains **three problems**. **Choose two of the three problems to solve.**
- Each problem is worth **50 points**. Your two selected problems therefore give a total of **100 points**.
- Clearly indicate which two problems you want to have graded. **Only two problems will be graded.**
- You do **not** receive additional credit for attempting all three problems.
- Show your reasoning and the important intermediate steps. An unsupported final answer may receive little or no credit.
- You may use a calculator.
- Any formulas or numerical information that are not expected to be known are provided in the individual problems.

Read all three problems before deciding which two you want to solve.

Problem 1 — Hyperfine Structure of Helium Ions

The two stable isotopes of helium have different nuclear spins:

- ^{3}He has nuclear spin $I = \frac{1}{2}$.
- ^{4}He has nuclear spin $I = 0$.

Consider the singly ionized atoms $^3\mathrm{He}^+$ and $^4\mathrm{He}^+$, each of which contains one electron. Their ground states are $1\,^2S_{1/2}$, with electron spin $S = \frac{1}{2}$ and orbital angular momentum $L = 0$.

(a) Hyperfine angular momentum — 10 points

The total atomic angular momentum is $\mathbf{F} = \mathbf{I} + \mathbf{S}$.

For each helium isotope, determine the possible values of F and the number of states in each hyperfine multiplet. Which isotope has a hyperfine splitting of its ground state?

(b) Hyperfine energies — 15 points

For an s electron, assume the hyperfine interaction is described by

$$\hat{H}_{\mathrm{HF}} = \frac{A}{\hbar^2}\hat{\mathbf{I}} \cdot \hat{\mathbf{S}},$$

where A is a constant with units of energy.

Starting from $\mathbf{F} = \mathbf{I} + \mathbf{S}$, express $\hat{\mathbf{I}} \cdot \hat{\mathbf{S}}$ in terms of the squared angular momenta. Then calculate the hyperfine energies of the $^3\mathrm{He}^+$ ground state in terms of A and determine their separation.

(c) Scaling with nuclear charge — 15 points

The normalized ground-state wavefunction of hydrogen is

$$\psi_{100}(r) = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0},$$

where

$$a_0 = \frac{4\pi\varepsilon_0\hbar^2}{m_e e^2}.$$

For a hydrogen-like ion with nuclear charge Z, the Coulomb potential is

$$V(r) = -\frac{Ze^2}{4\pi\varepsilon_0 r}.$$

1. How does the characteristic length scale (Bohr radius) change with Z? Determine a_Z in terms of a_0.
2. Write down the normalized ground-state wavefunction for a hydrogen-like ion with nuclear charge Z.
3. Determine how the probability density at the nucleus, $|\psi_{100}(0)|^2$, scales with Z.

You do not need to solve the Schrödinger equation.

(d) Hyperfine splitting in helium-3 — 10 points

The hyperfine interaction is proportional to the electron probability density at the nucleus,

$$A \propto g_N |\psi_{100}(0)|^2,$$

where g_N is the nuclear g-factor. Assume

$$g_{^3\mathrm{He}} = -g_p.$$

1. By what factor does the hyperfine splitting of $^3\mathrm{He}^+$ differ from that of hydrogen? Neglect reduced-mass corrections.
2. What is the consequence of the opposite signs of the nuclear g-factors for the ordering of the hyperfine energy levels?

Problem 2 — Electric-Dipole Transitions and Term Symbols

In LS coupling, an atomic state is labeled by the term symbol

$$n\,^{2S+1}L_J,$$

where S is the total spin, L is the total orbital angular momentum, and J is the total electronic angular momentum.

For electric-dipole (E1) transitions, use the selection rules

$$\Delta L = \pm 1, \qquad \Delta S = 0, \qquad \Delta J = 0, \pm 1,$$

with transitions $J = 0 \to J = 0$ forbidden.

(a) Allowed and forbidden transitions — 25 points

For each transition below, decide whether it is electric-dipole allowed. If it is forbidden, identify **all** violated selection rules. For this part, apply the selection rules to the stated quantum numbers, even if a term symbol is invalid.

1. $3\,^2D_{5/2} \to 2\,^2P_{3/2}$
2. $4\,^2D_{3/2} \to 2\,^2S_{1/2}$
3. $3\,^3P_1 \to 2\,^1S_0$
4. $5\,^4S_{1/2} \to 3\,^4P_{5/2}$
5. $4\,^2P_{1/2} \to 2\,^2S_{1/2}$

(b) Physically possible term symbols — 10 points

One of the term symbols in part (a) is not physically possible because its values of L, S, and J cannot be coupled to form that state.

Identify the invalid term symbol and explain your answer using the allowed values of J.

(c) Electric-dipole matrix element and symmetry — 15 points

Consider an excited hydrogen state with spatial wavefunction

$$\psi_e(\vec{r}) = Nxf(r),$$

where N is a normalization constant and $f(r)$ depends only on the radial coordinate r. The ground-state wavefunction $\psi_g(r)$ also depends only on r.

The electric-dipole transition matrix element is

$$\vec{d} = -e\langle g|\vec{r}|e\rangle.$$

1. Write down the integrands for the three Cartesian components d_x, d_y, and d_z, ignoring constant factors.
2. Using the symmetry of the integrands, determine which components of the electric-dipole matrix element vanish and which component can be nonzero. Briefly explain your answer.

You do **not** need to evaluate any integrals.

Problem 3 — Lithium Spectroscopy

In lithium, the transition

$$2\,^2P_{3/2} \to 2\,^2S_{1/2}$$

has an angular frequency of approximately

$$\omega_0 = 2.81 \times 10^{15}\,\mathrm{s}^{-1}.$$

The Doppler linewidth (FWHM) is given by

$$\delta\omega_D = \omega_0 \sqrt{\frac{8k_B T \ln 2}{mc^2}},$$

where m is the atomic mass, T is the temperature, k_B is Boltzmann's constant, and c is the speed of light.

(a) Isotope dependence of Doppler broadening — 15 points

Consider the two stable lithium isotopes, ^{6}Li and ^{7}Li, with approximate masses of $6u$ and $7u$, respectively, where u is the atomic mass unit.

1. Which isotope has the smaller Doppler linewidth at the same temperature?
2. By what factor do the Doppler linewidths of the two isotopes differ? Explain your reasoning.

(b) Doppler broadening and hyperfine splitting — 20 points

The ground state of ^{6}Li has a hyperfine splitting of

$$\omega_{\mathrm{hf}} = 2\pi \times 228 \times 10^6\,\mathrm{s}^{-1}.$$

Determine the temperature T at which the Doppler linewidth equals the hyperfine splitting,

$$\delta\omega_D = \omega_{\mathrm{hf}}.$$

You may leave your answer as a symbolic expression.

(c) Many-electron structure of lithium — 15 points

Lithium has three electrons and a nuclear charge of $Z = 3$.

1. Write down the ground-state electron configuration of lithium.
2. In the nonrelativistic hydrogen atom, the $2s$ and $2p$ orbitals have the same energy. Explain why this degeneracy is lifted in lithium.
3. Which orbital, $2s$ or $2p$, experiences the larger effective nuclear charge in lithium? Explain your answer in terms of electron screening and the spatial distribution of the orbitals.